Huawei Access Network Equipment Hardware/Environmental Fault Resolution SOP for WFD-ACC-09 at Whitefield POP

Node: WFD-ACC-09 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_8516

Description

The Huawei Access Network Equipment Hardware/Environmental Fault failure domain occurs on the Access Role due to ASIC memory parity errors, line card crashes, or power supply redundancy loss at the Whitefield POP. This is attributed to a combination of hardware and environmental factors, primarily the failure of redundant components, which are critical in maintaining network uptime and preventing Single Point of Failure (SPOF) scenarios. The Huawei eNodeB hardware architecture incorporates redundant components to ensure high availability and minimize downtime. However, in the event of a hardware or environmental failure, the node's operational integrity is compromised, resulting in the ALARM_8516 being triggered. This SOP outlines the steps to identify and resolve the root cause of the failure, restore network services, and ensure the node's operational integrity is maintained.

Pre-requisite Checks

- Verify the Whitefield POP is accessible and safe for remote hands to dispatch to the WFD-ACC-09 site.

- Check the status of redundant links and confirm they are functioning correctly.

- Review the recent maintenance history of the WFD-ACC-09 site to identify any recent changes or interventions.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display cpu-resource, Expected Output: CPU Resource Information: ..., Description: This command will provide detailed information about the CPU resources, including usage levels, which can help isolate the root cause of the failure.

Command: display board info, Expected Output: Board Information: ..., Description: This command will provide detailed information about the board's hardware components, including the status of the redundant components.
```

```
Command: display power supply, Expected Output: Power Supply
Information: ..., Description: This command will provide detailed
information about the power supply's status, including the presence of
redundancy.
```

Resolution Steps

Step 1: Step: Check the CPU resources and identify any anomalies., Command: display cpu-resource, Description: This step will provide critical information about the CPU resources, including usage levels, which can help identify potential issues.

Step 2: Step: Verify the board's hardware components and identify any discrepancies., Command: display board info, Description: This step will provide detailed information about the board's hardware components, including the status of the redundant components.

Step 3: Step: Verify the power supply's status and confirm the presence of redundancy., Command: display power supply, Description: This step will provide detailed information about the power supply's status, including the presence of redundancy.

Post-Check Verification

```
Command: display board info, Expected Output: Board Information: ...,
Description: This command will provide detailed information about the
board's hardware components, including the status of the redundant
components.

Command: display power supply, Expected Output: Power Supply
Information: ..., Description: This command will provide detailed
information about the power supply's status, including the presence of
redundancy.
```

Roll-back Procedures

- Step: Restore the original configuration., Command: system-view undo load command, Description: This step will restore the original configuration, effectively rolling back the changes made during the resolution process.

- Step: Reboot the node., Command: reboot, Description: This step will restart the node, which may resolve any issues caused by the resolution process.